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Roll No

EX-502 (GS)
B.E. V Semester Examination, June 2020
Grading System (GS)
Microprocessor and Microcontroller
Time : Three Hours

Maximum Marks : 70

Note: i) Attempt any five questions.

ii) All questions carry equal marks.

1. Write a program to generate square wave using programmable peripheral interface chip.
2. Write an algorithm and ALP for 8086 microprocessor to subtract two 8 bit ASCII members.
3. a) Write a short note on USART 8251.
b) Comment on instruction set of 8051.
4. Write an assembly language program using 8051 instruction set to generate a time delay of 0.2msec.
Assume that XTAL=11.0592MHz.
5. a) Explain how a timer is programmed in 8051.
b) Explain the interfacing of stepper motor to 8051 microcontroller.
6. Write short notes on any two
 - i) Serial port programming
 - ii) 8051 based thyristor firing circuit
 - iii) Programmable Internal Timer(8254)
 - iv) Subroutine calls and returns.
7. What is the difference between microprocessor and microcontroller? Draw the pin diagram of 8051 microcontroller.
8. Write short notes on any two:
 - a) Timing diagram
 - b) PPI 8255 Architecture
 - c) Pin configuration of 8086
 - d) Serial communication modes
